Humanvitha Chinnam

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Summary

Quality Engineer with 3+ years of combined software development and quality engineering experience across enterprise banking and FinTech. Experienced in building Python and VBA automation tools, internal quality tooling, and scalable test solutions for complex financial systems. Strong software engineering foundation across Python, JavaScript/TypeScript, React, APIs, test automation, and CI/CD, with experience translating complex business requirements into reliable automated validation workflows.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C, SQL
Quality Engineering: Test Automation, Functional Testing, Regression Testing, Integration Testing, End-to-End Testing, Test Planning, Defect Analysis, Root Cause Analysis
Automation & Tooling: Playwright, Python Scripting, VBA, Jest, API Testing, Postman, TSYS/Mainframe Testing
Web & APIs: React.js, Next.js, Angular, HTML, CSS, Tailwind CSS, Zustand, Drupal, Node.js, Django, GraphQL
Databases & Cloud: MySQL, MongoDB, DynamoDB, Oracle, AWS, GCP, React Query
Data & ML: Pandas, NumPy, Matplotlib, Scikit-learn, OpenCV, Tableau, Jupyter Notebook
Tools & Practices: GitHub, GitLab, Gerrit, Jira, Confluence, VS Code, Figma, Docker, Postman

Experience

Quality Engineer Jul 2025 – Present
Barclays, USA

- Architected and deployed a VBA-based automation platform interfacing directly with the TSYS mainframe for end-to-end credit card change validation — reduced a 2-day manual testing cycle to 45 minutes (∼95% time reduction), saving 200+ engineering hours per quarter and accelerating release velocity across the team.
- Built internal automation tooling using Python and VBA to streamline end-to-end validation of complex credit-card workflows on TSYS, reducing repetitive manual testing and accelerating regression cycles.
- Engineered a Python-based AI computer vision pipeline leveraging image differencing, OCR, and ML-based pattern recognition to automate validation of customer credit card packet documents — achieved 99.2% defect detection accuracy across text, images, and barcodes, eliminating manual visual review entirely.
- Built an AI-driven rewards verification engine that programmatically validates reward point assignments across 50,000+ transactions per cycle against complex business rules — reduced reward calculation errors by 95% and eliminated error-prone manual spot-checking.
- Designed and developed intelligent TSYS script analysis tools to compare large-scale production scripts impacting customer-facing credit card workflows — improved change visibility by 80%, reduced release-related production incidents by 40%, and minimized risk across thousands of lines of production code.
- Led end-to-end testing strategy for credit card management workflows on the TSYS mainframe covering transaction posting, rewards accrual, statement generation, and UI-layer validation — achieved 99.5% defect escape rate reduction across 12+ production releases while establishing CI/CD-integrated test gates with AI-powered anomaly detection to flag regression risks early.
- Developed predictive test prioritization logic using historical defect data and AI-based risk scoring to optimize test execution order — reduced regression cycle time by 35% while maintaining full coverage, and reduced post-deployment hotfixes.
- Partnered with product, engineering, and compliance teams to translate complex credit card rules and regulatory requirements into structured, auditable automated test scenarios — improved test coverage by 60%, strengthened compliance audit readiness, and ensured correctness across reward, transaction, and fraud-detection flows.

Software Engineer Intern Jun 2024 – Dec 2024
WalletGyde, Colorado

- Migrated the platform end-to-end from Bubble.io to a production-grade Next.js and DynamoDB application, delivering fully responsive desktop and mobile interfaces that contributed to a 40% increase in user adoption.
- Created a Personal Loan Calculator landing page with real-time estimation logic, directly generating 300+ new customer leads.
- Implemented REST APIs that reduced data retrieval latency by 15%, measurably improving system responsiveness across the platform.
- Researched AI-based matchmaking algorithms, improving successful match rates by 25% over legacy approaches.

Graduate Assistant – Data & Web Systems Mar 2024 – May 2025
CCPHIT, California State University, Long Beach

- Restructured and standardized dataset management processes to ensure data integrity across research workflows, achieving 99% data accuracy and supporting more reliable academic decision-making.
- Modernized a Drupal-based CMS to streamline updates for course offerings and internship opportunities and automate outreach — resulting in a 25% increase in student enrollments and internship placements.

Front-End Developer Jul 2021 – Aug 2023
Tata Consultancy Services (TCS), India

- Crafted responsive React.js web interfaces for Lloyds Bank's enterprise banking platform, improving performance and driving a 30% increase in client satisfaction scores
- Automated a regulatory-update workflow on the Lloyds banking platform, eliminating 60% of manual processing time and enabling the client to meet compliance deadlines consistently.
- Authored a reusable React component library adopted across multiple Lloyds product lines, accelerating team development velocity by 25% while maintaining scalability and maintainability.
- Optimized code review and release workflows using Gerrit and GitHub, cutting the review cycle by 40% and shortening project delivery timelines by 30%.
- Integrated RESTful APIs with backend teams to power real-time data flow across customer-facing banking workflows, improving reliability and reducing data-related defects.
- Ensured WCAG 2.1 AA accessibility compliance across Lloyds banking interfaces, implementing semantic markup, keyboard navigation, and screen-reader support to meet UK financial accessibility standards.
- Raised Jest test coverage through component-level testing, reducing production bugs by 40% and improving overall release quality.
- Validated cross-browser and responsive behavior across desktop, tablet, and mobile breakpoints, ensuring consistent rendering and functionality for a broad customer base.
- Improved frontend performance through lazy loading, code splitting, and bundle-size optimization, strengthening Core Web Vitals and reducing initial page load times.
- Contributed actively to Agile ceremonies—sprint planning, stand-ups, and retrospectives—supporting consistent on-time delivery across release cycles.

Projects

HearHub Headphones – E-commerce Platform
React.js, Sanity, Tailwind CSS, Stripe

- Launched a full-stack e-commerce platform with Stripe payment processing and a Sanity CMS catalog, boosting user retention by 40% and reducing payment failures by 60%.

Reliever – Social Media Application
Next.js, TypeScript, GraphQL, Zustand

- Architected a video-sharing platform with optimized media caching and GraphQL data fetching, cutting load times by 40% and improving search performance by 25%.

Real-Time Chat Application
React.js, JavaScript, Chat Engine, WebSockets

- Engineered a live messaging application using WebSockets for bidirectional communication, slashing latency by 50% and increasing user engagement by 30%.

Education

California State University, Long Beach Aug 2023 – May 2025
Master of Science in Computer Science (GPA: 3.7)

Lakireddy Bali Reddy College of Engineering Aug 2017 – Jul 2021
Bachelor of Technology in Computer Science (CGPA: 9.28)

Certifications & Publications

Google Cloud Certified Associate Cloud Engineer

Real-Time Face Mask Detection System
OpenCV, DNN, Transfer Learning ***(95% accuracy)***